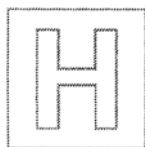


NAME: _____ CLASS: _____ INDEX: _____



CATHOLIC JUNIOR COLLEGE
JC2 MIDYEAR EXAMINATIONS
Higher 2

BIOLOGY

Paper 1 Multiple Choice

9648/01

MONDAY 19 SEPTEMBER 2011

1 Hour 15 Minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write and/or shade your name, NRIC / FIN number and HT group on the Answer Sheet in the spaces provided unless this has been done for you.

There are **40 MCQ** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **2B pencil** on the separate Answer Sheet.

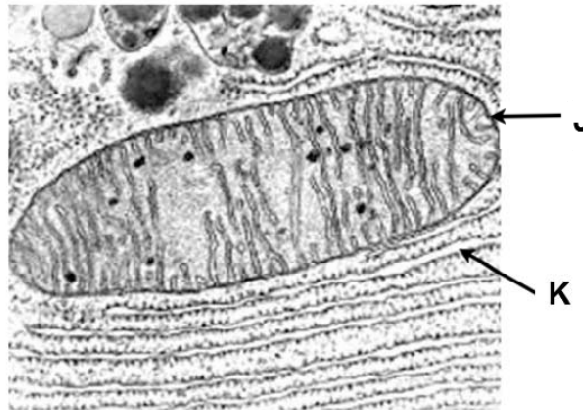
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

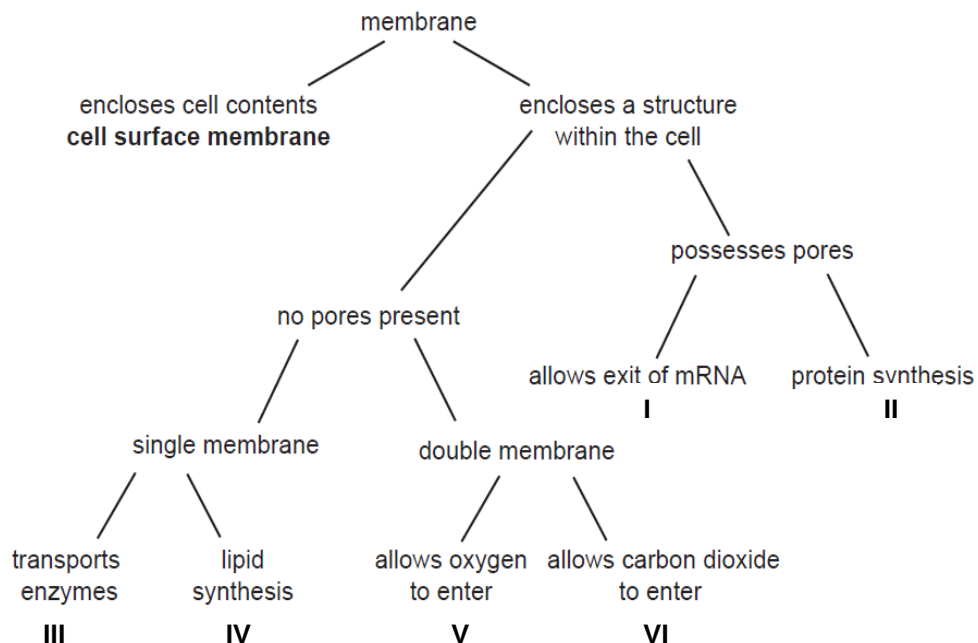
Calculators may be used.

1. The electronmicrograph below shows the contents of a cell.



Which of the following describes a possible functional relationship between **J** and **K**?

- A** Polypeptides synthesised in **J** are transported to **K** for further modification.
 - B** Polypeptides synthesised in **K** are transported to **J** for further modification.
 - C** ATP synthesised in **J** is directed towards polymerisation of amino acids at **K**.
 - D** mRNA is first synthesised in **J** and subsequently undergoes translation at **K**.
2. Membranes within and at the surface of cells have different roles. The flowchart shows a scheme to identify various organelles within the cell, by describing the structure and function of their membranes.



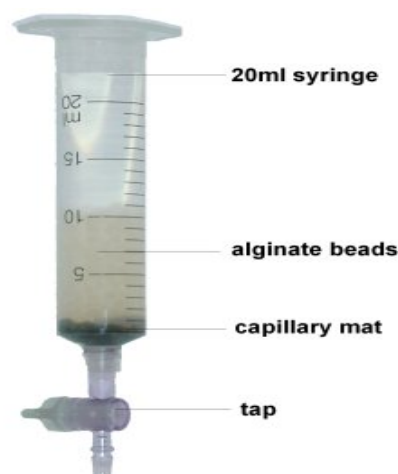
Which of the following outcomes correctly identifies the various cellular organelles?

	I	II	III	IV	V	VI
A	chloroplast	vesicle	smooth ER	rough ER	nucleolus	mitochondrion
B	nucleus	smooth ER	mitochondrion	rough ER	vesicle	chloroplast
C	nucleolus	rough ER	vesicle	smooth ER	nucleus	mitochondrion
D	nucleus	rough ER	vesicle	smooth ER	mitochondrion	chloroplast

3

3. Molecules of a stomach enzyme were immobilised (strongly attached) in alginate beads and used in an investigation to determine the identity of four different unknown substances (W, X, Y, Z) that affected the enzyme activity even when its substrate was present in excess.

The beads were packed into a column set up similar to the one shown below.



In order to allow the enzyme activity to equilibrate, 10ml of substrate solution was poured in from the top and allowed to drain through the alginate bead column thrice before the experiment was started.

Following the equilibration, 10ml of unknown substance W was allowed to drain through the alginate bead column fully before another 10ml of substrate solution was drained through the column to allow the enzymes to react with the substrates. The concentration of product obtained in the filtrate was measured to reflect the level of enzyme activity after addition of the unknown substance. These steps were then repeated with the other three unknown substances added into the same alginate bead column in the order of X, Y, then Z.

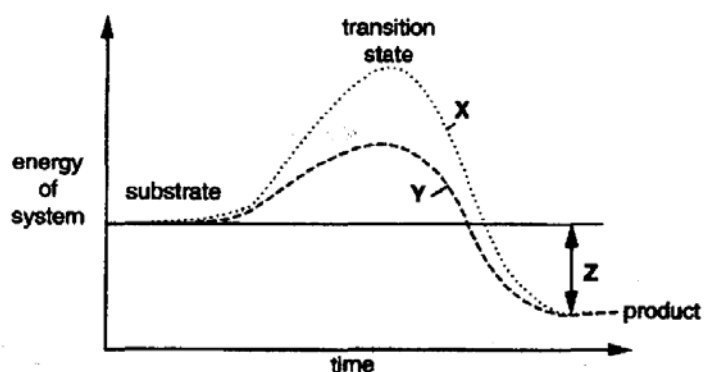
The table below shows the results obtained.

Unknown substance	Enzyme activity after addition of unknown substance (arbitrary units)
W	20.7
X	8.5
Y	0.1
Z	0.1

Given that the enzyme activity in the absence of any unknown substance was 16.8 arbitrary units after equilibration, which combination correctly identifies the four unknown substances based on the results obtained?

	W	X	Y	Z
A	cofactor	pH buffer at pH 2	competitive inhibitor	non-competitive inhibitor
B	substrate	cofactor	pH buffer at pH 7	competitive inhibitor
C	cofactor	non-competitive inhibitor	pH buffer at pH 13	substrate
D	substrate	competitive inhibitor	non-competitive inhibitor	cofactor

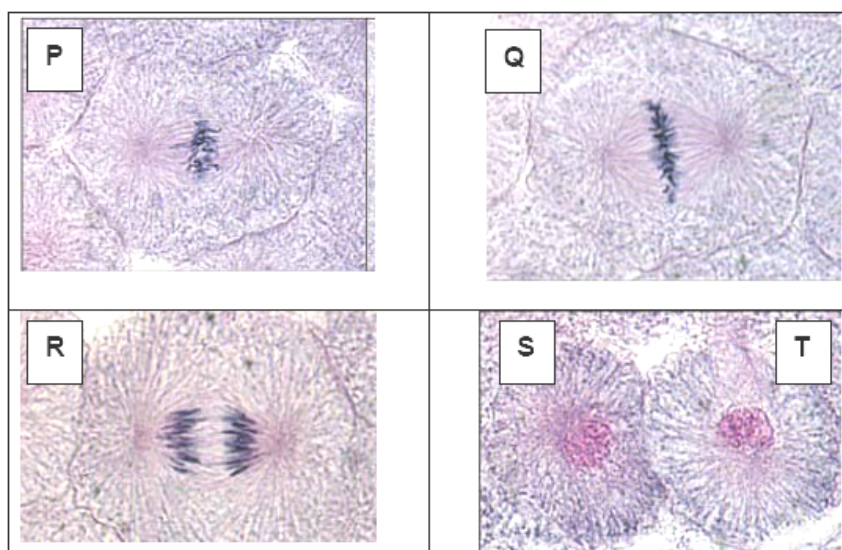
4. The graph shows the effect of an enzyme on a reaction.



Which combination identifies X, Y and Z?

	X	Y	Z
A	catalysed reaction	uncatalysed reaction	activation energy
B	catalysed reaction	uncatalysed reaction	energy lost during reaction
C	uncatalysed reaction	catalysed reaction	energy lost by product
D	uncatalysed reaction	catalysed reaction	overall energy change

5. Cells P, Q and R are undergoing various stages of mitosis. Cells S and T are daughter cells formed immediately after cell division.



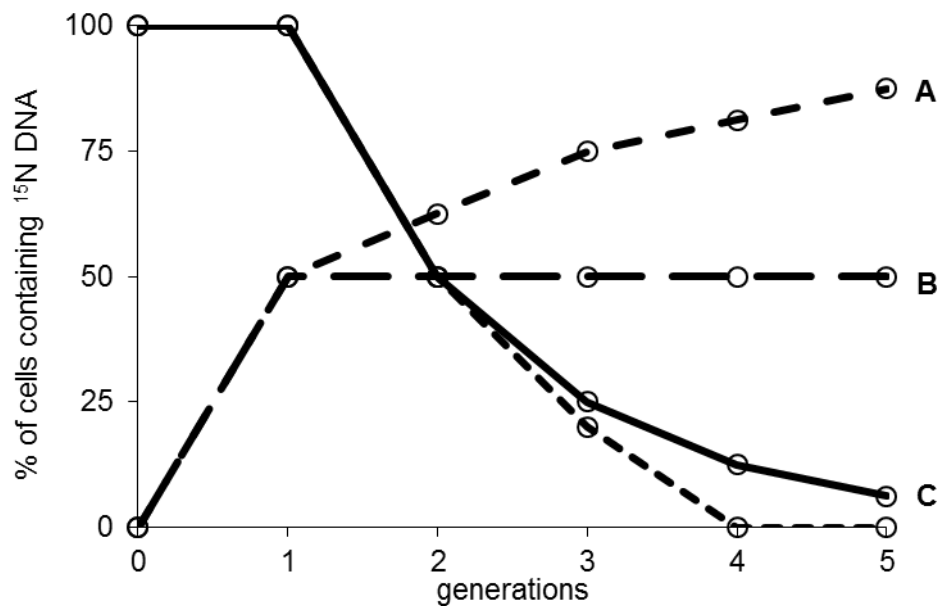
The plant tissue had been growing in a culture medium containing radioactive DNA nucleotides. The level of radioactivity in each cell was measured and the presence of sister chromatids were noted. Cells with the same amount of DNA have approximately the same level of radioactivity.

Which cell has a level of radioactivity and observations that could have been resulted due to inaccurate readings and/or observations?

	Cell	Radioactivity level (a.u.)	Presence of sister chromatids
A	P	102	Yes
B	Q	101	Yes
C	R	52	No
D	S	51	No

6. One complete turn of the double helix of DNA contains 10 pairs of bases and is 3.4 nm long. What is the approximate length of the DNA coding sequence of lysozyme, a protein of 129 amino acids?
- A 70 nm
B 132 nm
C 66 nm
D 140 nm
7. Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all the DNA was labelled. These bacteria were then grown in a medium containing only normal nitrogen (^{14}N) for five generations. The percentage of cells containing ^{15}N in each generation was estimated.

Which curve provides evidence that DNA replication is semi-conservative?



8. What prevents simultaneous transcription and translation in eukaryotes?
- A The lack of gene clusters into operons
B The absence of three different RNA polymerases
C The requirement to remove exons
D The nuclear membrane around the DNA
9. Which of the following statement is true of the end-replication problem of DNA?
1. When a linear DNA molecule replicates, a gap is left at the 3' end of each new strand because DNA polymerase can only add nucleotides to a 5' end.
 2. Telomerase prevents the end-replication problem from occurring.
 3. Repeated rounds of replication produce shorter and shorter DNA molecules.
 4. Prokaryotes do not have the end-replication problem.
- A 1 and 2
B 1 and 4
C 2 and 3
D 3 and 4

10. The following events occur in the extension of telomeres.

1. Further extension of 3' end of telomere.
2. Translocation of telomerase to end of telomere.
3. Reverse transcription to extend the 3' end of the telomere.
4. Complementary base-pairing of RNA template with single-stranded end of telomere.

In which order do these events take place?

- A 3 2 4 1
- B 3 4 2 1
- C 4 2 3 1
- D 4 3 2 1

11. Rhabdoviruses infect human cells. The genome of rhabdoviruses consists of a single-stranded RNA molecule whose sequence is complementary to the RNA sequence which functions as a messenger RNA. How is the "+" messenger RNA produced in such cells by rhabdovirus?

- A Host cell RNA polymerase activity.
- B One portion of the infecting RNA is directly translated by host cell ribosomes.
- C The infecting virus particle contains an RNA-dependent RNA polymerase.
- D Reverse transcriptase activity.

12. Which of the following statements about the life cycle of the Human Immunodeficiency Virus are *false*?

1. The genome of HIV consists of 2 copies of linear, single-stranded RNA enclosed in a icosahedral capsid.
2. The viral DNA which enters the host cell's nucleus will be integrated into the genetic material of the host cell using the host cell's enzyme, integrase.
3. The HIV particles recognise the host cell through the sialic-acid containing proteins or lipids on the membrane of the host cell.
4. The viruses are released from the host cell by exocytosis.

- A 1 and 3 only
- B 2 and 4 only
- C 2, 3 and 4 only
- D All of the above

13. In generalised transduction, defective viruses are formed as a result of

- A viral enzymes cutting the host DNA such that the host DNA is assembled into the new virus.
- B production of host enzymes by virus which nicks its own DNA such that it can be assembled into the new virus.
- C no shut down of host DNA production such that either the host DNA or the virus DNA can be assembled into the new virus.
- D integration of viral DNA into host DNA and during excision of the prophage, the viral genome carries along with it host DNA to be assembled into the new virus.

14. *Staphylococcus aureus* is a common bacterium, found on human skin. There are many strains of *S. aureus*. The antibiotic methicillin was then used to treat infection by *S. aureus*. Now there are at least 15 different strains of MRSA (methicillin resistant *Staphylococcus aureus*).

Which of the following are valid reasons for the emergence of 15 different strains of MRSA?

- I The bacteria mutated when it was exposed to methicillin, thus becoming resistant.
- II The bacteria underwent spontaneous mutation and some strains happened to be resistant to methicillin.
- III The bacteria may have been exposed to a virus that carried a methicillin-resistant gene and underwent specialized transduction.
- IV The antibiotic caused the bacteria to produce methicillin-resistant proteins.

- A I and II
- B II and III
- C II, III and IV
- D I, III and IV

15. Which of the following is true of gene regulation in both prokaryotes and eukaryotes?

- A Gene regulation of eukaryotes involves bending of DNA while in prokaryotes, chemical modifications to DNA is required.
- B In eukaryotes, increase in efficiency of RNA polymerase requires specific transcription factors to activate the initiation complex while in prokaryotes, activators are required to bind directly to promoter site to increase efficiency of RNA polymerase.
- C In eukaryotes, RNA polymerase binds to the TATA box while in prokaryotes, RNA polymerase binds to the regulatory gene.
- D Repressors and activators regulate expression in eukaryotes while prokaryotes are regulated by repressors only.

16. In *E. coli*, the production of enzymes for tryptophan synthesis is carefully controlled according to the organism's metabolic needs. A mutation in the gene encoding the tryptophan repressor has occurred, such that the repressor can bind DNA without the co-repressor.

What effect on enzyme production can be expected under this condition?

- A High-level enzyme production under any condition
- B High-level enzyme production in the absence of tryptophan, no activity in the presence of tryptophan
- C No enzyme production in the absence of tryptophan, high-level activity in the presence of tryptophan
- D No enzyme production under any conditions

17. Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein it codes for?

	Promoter	Transcription terminator	Start codon	Stop codon	Middle of an intron
A	No protein product is produced	Protein product is shorter than normal	Protein product is longer than normal	Protein product is normal	Too much protein product is produced
B	Too much protein product is produced	Protein product is normal	No protein product is produced	Protein product is longer than normal	Protein product is normal
C	Protein product is normal	Protein product is longer than normal	Protein product is shorter than normal	Too much protein product is produced	Protein product is longer than normal
D	Protein product is longer than normal	Too much protein product is produced	Protein product is normal	Protein product is shorter than normal	No protein product is produced

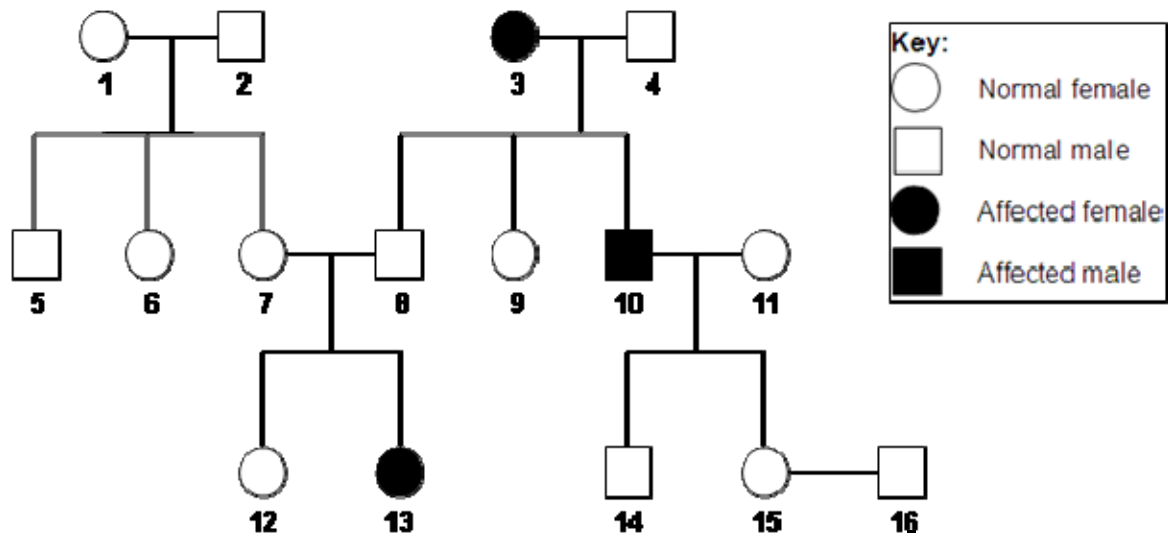
18. Which statement about cancer is not correct?

- A** Both alleles for a tumour suppressor gene must undergo loss-of-function mutations in the development of cancer.
- B** Mutations in proto-oncogenes may not necessarily result in excessive cell proliferation.
- C** The growth of new blood vessels is a fundamental step in the development of a malignant tumour.
- D** The risk of cancer is higher when *p53* gene is placed under the control of a strong viral promoter.

19. A cross was performed between two organisms with genotype AABBCc and AabbCc. Assuming complete dominance and independent assortment, what proportion of the progeny will be expected to phenotypically resemble the first parent?

- A** 1/4
- B** 1/16
- C** 3/4
- D** 9/16

20. The pedigree shows the inheritance of a condition known as galactosemia.



What is the probability that the first child of individuals **15** and **16** will be a normal boy?

- A $3/4$
 B $3/8$
 C $7/8$
 D $7/16$
21. In pea plants, the allele (Y) coding yellow seeds is dominant to the allele (y) coding green seeds. The allele (R) coding for round seeds is dominant to the allele (r) coding for wrinkled seeds. Plants grown from yellow, round seeds were crossed with plants grown from green, wrinkled seeds.

The table below shows the phenotype of 80 offspring plants.

Number of offspring	Seed colour	Seed shape
23	yellow	round
20	yellow	wrinkled
17	green	round
20	green	wrinkled

Which of the following sets of parental genotypes would produce the observed results above?

- A $YYRR \times yyrr$
 B $YyRR \times yyrr$
 C $YyRr \times yyrr$
 D $YyRr \times YyRr$

22. In the broad bean, a pure-breeding variety with green seeds and black hilums (the point of attached of the seed to the pod) was crossed with a pure-breeding variety with yellow seeds and white hilums. All the F_1 plants had yellow seeds and white hilums. When these were allowed to self-fertilise, the plants of the F_2 generation produced the following seeds.

Yellow seeds and white hilums	89
Yellow seeds and black hilums	28
Green seeds and white hilums	25
Green seeds and black hilums	45

A chi-squared (χ^2) test was performed to test the significance of the difference between the observed and expected results.

$$\chi^2 = \sum \frac{(O-E)^2}{E} \quad v = c - 1$$

degrees of freedom	$p = 0.5$	$p = 0.1$	$p = 0.05$	$p = 0.01$	$p = 0.001$
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.36	7.78	9.49	13.28	18.46

Which combination correctly describes the results of the χ^2 test?

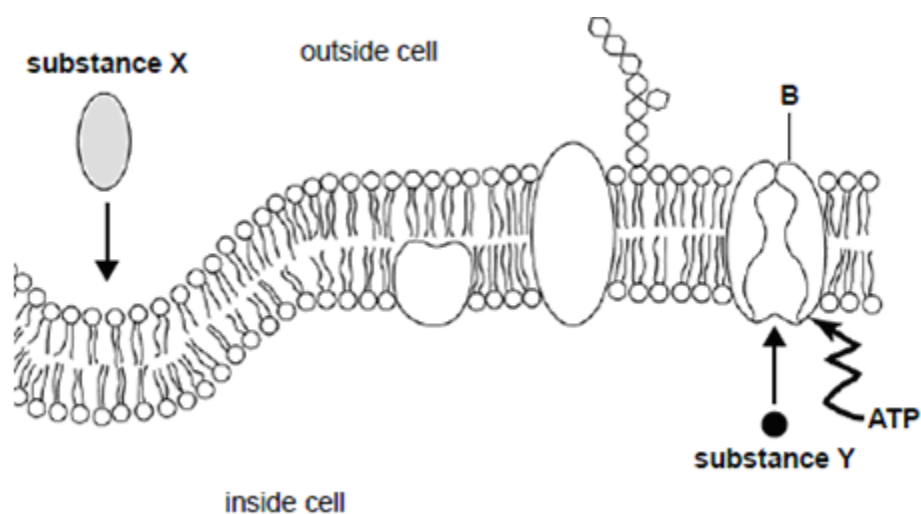
	number of degrees of freedom	probability	these are two pairs of segregating alleles at two loci
A	3	> 0.05	yes
B	3	< 0.05	no
C	4	> 0.05	yes
D	4	< 0.05	no

23. Which of the following statements show the advantages of signal transduction pathways?

1. Amplification of signals.
2. Fine-tuning of cellular responses.
3. Different responses from a single cell.
4. Switch on certain genes while other genes are switched off.

- A 1 only
 B 1 and 2
 C 2, 3 and 4
 D 1, 2, 3 and 4

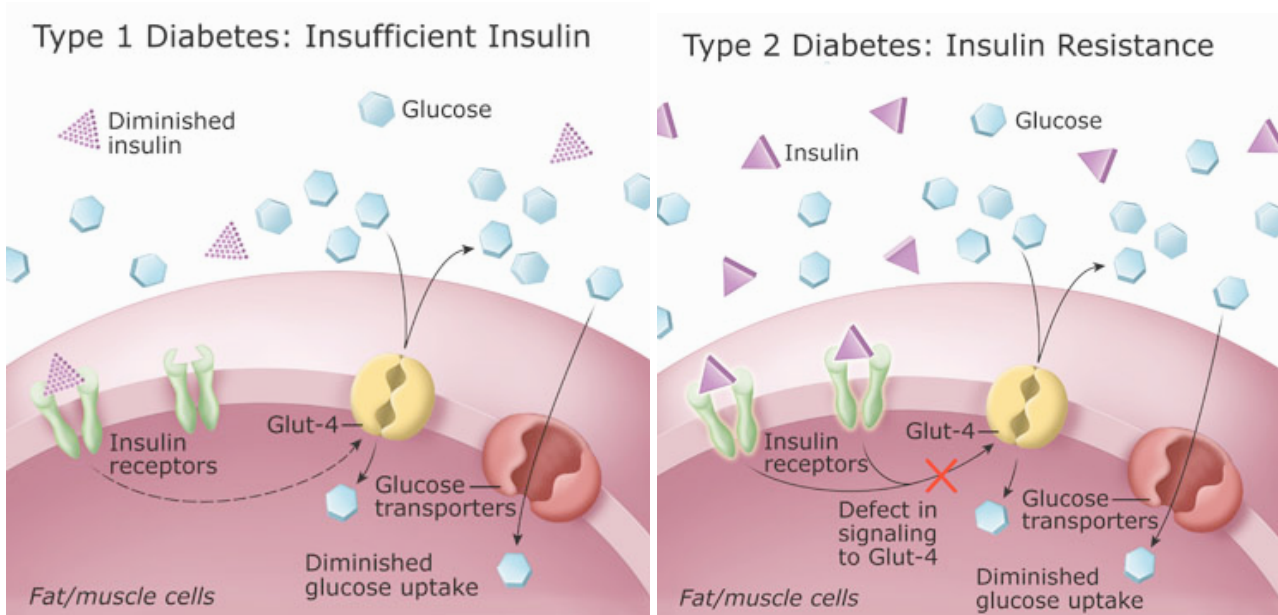
24. The figure below shows a cross section of the plasma membrane in a typical mammalian cell. The substances labelled **X** and **Y** are about to be transported across the membrane in the directions shown by the arrows ($\longrightarrow$).



Which of the following is correct?

	X			Y		
	<i>type of movement</i>	<i>nature of molecule</i>	<i>size of molecule</i>	<i>type of movement</i>	<i>nature of molecule</i>	<i>size of molecule</i>
A	diffusion	lipid soluble	small	facilitated diffusion	lipid soluble	large
B	diffusion	water soluble	small	active transport	lipid soluble	small
C	endocytosis	lipid soluble	large	facilitated diffusion	charged	large
D	endocytosis	water soluble	large	active transport	charged	small

25. The following diagrams show the effects of Type 1 and Type 2 diabetes. In Type 1, insulin production is diminished and in Type 2, sensitivity of the insulin receptor is diminished.



From the diagrams, the deficiencies in the homeostatic feedback mechanism for the respective types of diabetes in glucose regulation are as follows:

I	Type 1	Effector cells normal	Receptor cells malfunction
II	Type 2	Effector cells normal	Receptor cells malfunction
III	Type 1	Beta-cells normal	Insulin receptors malfunction
IV	Type 2	Beta-cells normal	Effector cells malfunction

Which of the above deficiencies in the homeostatic feedback mechanism have been correctly described?

- A I and IV only
- B I and II only
- C II and III only
- D III and IV only
26. Dinitrophenol is a metabolic poison that can lodge within the thylakoid membranes of chloroplasts. It then provides an alternate route for H^+ ions to diffuse across the thylakoid membranes. In what way will the Calvin cycle (dark reaction) be affected in chloroplasts poisoned with dinitrophenol?
- A No effect since Calvin cycle is an enzyme-controlled process.
- B The rate of Calvin cycle will increase because pH in the stroma will drop towards the optimum.
- C The rate of Calvin cycle will decrease with the accumulation of glycerate-3-phosphate (GP).
- D The rate of Calvin cycle will decrease with the accumulation of triose phosphate (TP).

27. Two test tubes containing the following contents are shown below:

Tube 1:

Radioactive glucose solution + yeast cells suspension + oxygen + antimycin

Tube 2:

Radioactive glucose solution + yeast cells suspension + oxygen

Radioactive glucose has all its six carbons made of radioactive ^{14}C . The initial radioactivity measured for the glucose in each test tube is 60 arbitrary units.

Antimycin is an electron transport chain inhibitor.

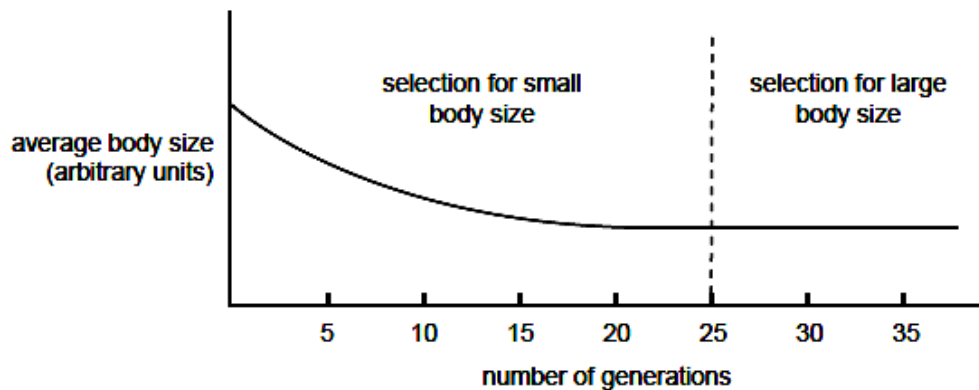
If the gaseous product and the aqueous products are tested using a radioactive meter after all the glucose has been metabolized, what would be the final observed readings?

	tube 1 (radioactivity measured/ arbitrary units)		tube 2 (radioactivity measured/ arbitrary units)	
	aqueous products	gaseous products	aqueous products	gaseous products
A	0	60	40	20
B	20	40	0	60
C	40	20	0	60
D	40	20	60	0

28. What is one similarity and one difference between classification and phylogeny?

- A** Both organise organisms into groups based on their characteristics but phylogeny takes into consideration the evolutionary relationships between groups.
- B** Both put organisms together according to increasingly inclusive groups but classification does not show a branching pattern of evolutionary relationships.
- C** Both rely on physical characteristics of organisms but phylogeny does not require scientific names.
- D** Both take into account the common ancestors of organisms but classification is hierarchical in nature.

29. The graph shows the results of a breeding experiment with the vinegar fly. In each of the first 25 generations the smallest flies were selected to produce the next generation. After 25 generations the selection was reversed. From generations 25 to 35, the largest flies were chosen to breed the next generation.



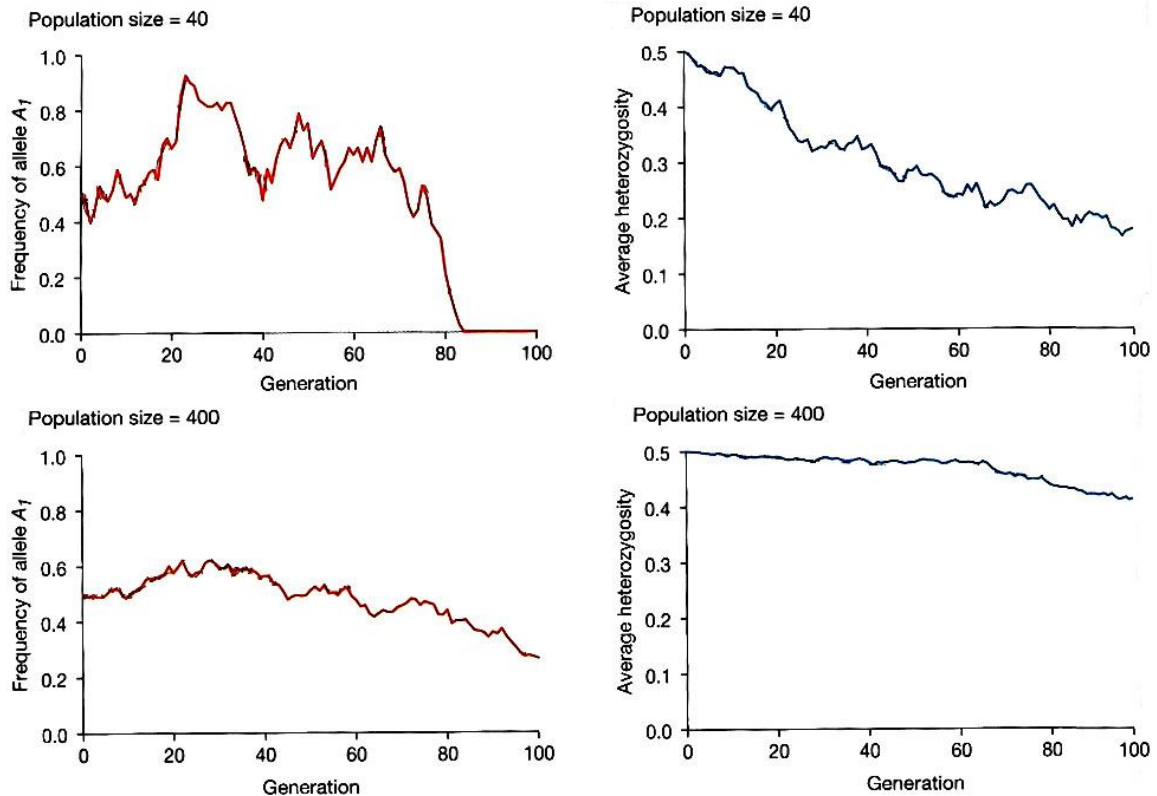
With respect to the genes controlling body size, the results of the experiment suggest that

- A after 25 generations, there was no genetic variation.
 B for the first 25 generations, there was no genetic variation.
 C selection between generations 25 and 35 had a significant effect on average body size.
 D if selection for small body size continued after generation 25, average body size would continue to fall.
30. Two well-known members of the cat genus *Panthera* are the lion (*P. leo*) and the tiger (*P. tigris*). Lions are highly social, living in large groups. Tigers are solitary animals except during the mating season. The natural ranges of the two species also have a small region of overlap. Two types of hybrids known as ligers (male lion × female tiger) and tigrlons (male tiger × female lion) have been created in captivity. Although F_1 hybrids of both types typically survive, F_2 hybrids are very rare.

Which of the following isolation mechanisms could account for the formation of these two species?

- I Geographical
 II Physiological
 III Behavioural
- A I only
 B I and III only
 C II and III only
 D I, II and III

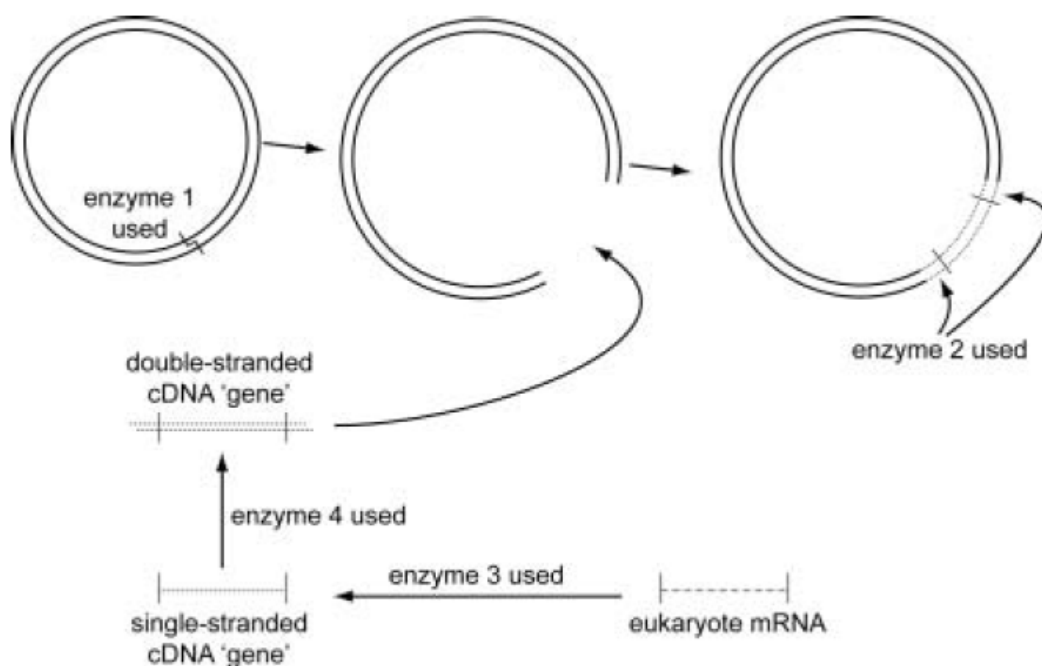
31. The four graphs represent simulations of genetic drift occurring in fruit fly populations of two different sizes, 40 and 400.



Which of the following about genetic drift can be deduced from the graphs?

- A It leads to non-random fixation of alleles in the population.
- B There is loss of heterozygosity in the population regardless of population size.
- C It is a more powerful force of evolution in small populations.
- D Given sufficient time, genetic drift can produce drastic changes in allele frequencies in large populations.

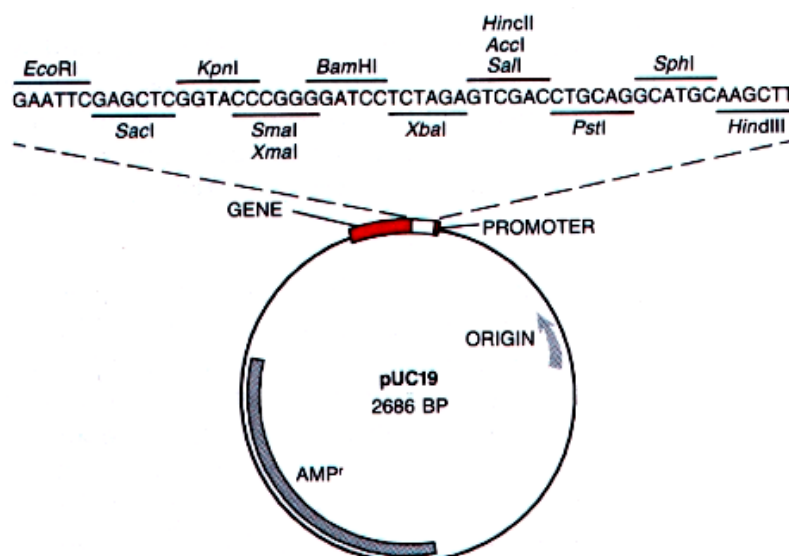
32. The diagram shows part of a process used for genetic engineering



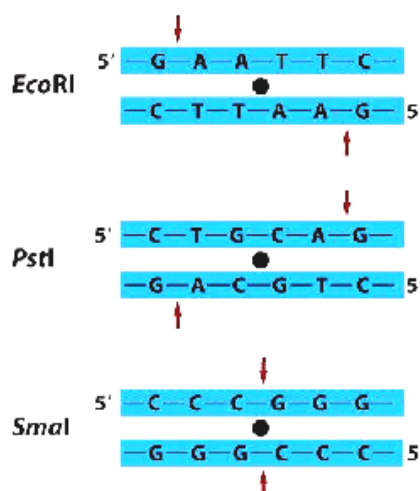
Which of the following correctly gives the identity of the enzymes used?

	Enzyme 1	Enzyme 2	Enzyme 3	Enzyme 4
A	Restriction enzyme	DNA ligase	Reverse transcriptase	Taq polymerase
B	DNA ligase	Restriction enzyme	DNA polymerase	Reverse transcriptase
C	Restriction enzyme	DNA ligase	Reverse transcriptase	DNA polymerase
D	DNA ligase	Restriction enzyme	Taq polymerase	Reverse transcriptase

33. The figure shows the plasmid map of pUC19, which includes the positions of the origin of replication (ORI), multiple cloning sites (MCS) and two genes - *amp^R* and *lacZ*. pUC19 is used for the production of human anti-thrombin III (hATIII) in the bacterium *E. coli*.



The restriction sites of three particular restriction enzymes, *EcoRI*, *PstI* and *SmaI* are shown below.



Which of the following can be concluded from the figures?

I	Any restriction enzyme whose restriction site is found within the MCS can be used to cleave pUC19.
II	pUC19 cleaved with <i>EcoRI</i> can anneal to hATIII cDNA cleaved with <i>PstI</i> as both restriction enzymes generate single-stranded overhangs of the same length.
III	The use of <i>SmaI</i> will require an additional step of ligating adaptors to the ends of pUC19 and hATIII cDNA.
IV	Colonies of <i>E. coli</i> transformed with recombinant pUC19 will survive on agar plates containing ampicillin and X-gal, giving a blue appearance.

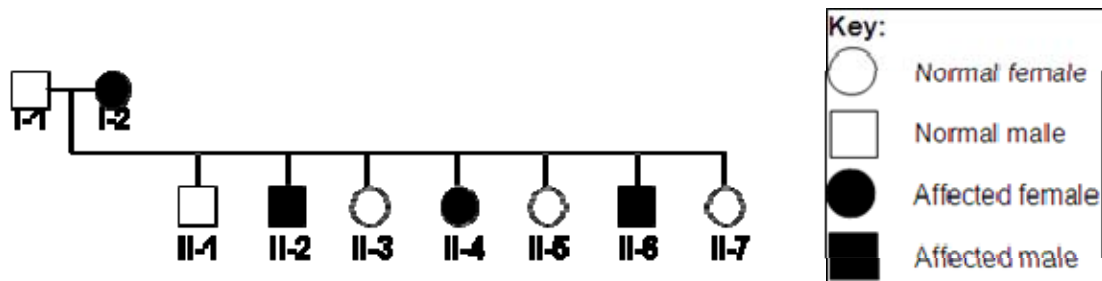
- A I and IV only
 B II and III only
 C I and III only
 D I, II, III and IV

34. How is the polymerase chain reaction (PCR) similar to the replication of DNA?

1. DNA is heated to break hydrogen bonds
2. DNA unzips
3. free nucleotides are used
4. DNA polymerase are required

- A 1 only
- B 2 and 4 only
- C 1, 2 and 3 only
- D 2, 3 and 4 only

35. Adult polycystic kidney disease (APKD) is inherited in an autosomal dominant manner. In an investigation to determine the chromosomal locus of APKD, linkage analysis of the APKD gene was carried out on members of one family. Three separate VNTR loci, **P**, **Q** and **R**, located on separate chromosomes, were used. The results of the linkage analysis are shown in the figure.



	Alleles present at various individuals' VNTR loci								
	I-1	I-2	II-1	II-2	II-3	II-4	II-5	II-6	II-7
VNTR locus P	2, 6	1, 6	1, 2	2, 6	1, 6	6, 6	1, 2	6, 6	1, 6
VNTR locus Q	1, 5	2, 7	7, 5	2, 5	7, 1	2, 1	7, 5	2, 1	7, 1
VNTR locus R	4, 8	5, 8	5, 8	8, 8	5, 4	8, 4	5, 8	8, 4	5, 4

Which of the following can be concluded from the figure?

- A Individual I-1 is unaffected as he does not possess two copies of allele 5 for VNTR locus Q
- B Individual II-2 is affected as he inherited two copies of allele 8 for VNTR locus R
- C Individual II-5 is unaffected as she did not inherit allele 2 for VNTR locus Q
- D Individual II-6 is affected as he inherited two copies of allele 6 for VNTR locus P

36. Non-viral *ex vivo* transfer of a gene encoding coagulation factor VIII was performed using stem cells isolated from the bone marrow of patients suffering from severe hemophilia A.

What is the sequence of events for the *ex vivo* transfer of the gene encoding coagulation factor VIII?

- I Transfection with plasmids containing the gene encoding coagulation factor VIII
- II Implantation of cells into patients
- III Isolation of stem cells
- IV Selection and cloning of cells expressing coagulation factor VIII

- A II, III, I, IV
 - B III, I, IV, II
 - C III, II, IV, I
 - D III, IV, II, I
37. Somatic line therapies are licensed for the treatment of genetic disorders rather than germ line therapies because
- A viruses can be used as vectors in somatic line therapies.
 - B somatic cell manipulation is more effective than germ cell manipulation.
 - C there is more control where the DNA inserts in somatic line therapies.
 - D the transformation is not carried on to the next generation in somatic line therapies.
38. In mice transfected with the rabbit β -globin gene, the rabbit gene is active in several tissues, including the spleen, brain, and kidney. In addition, some mice suffer from thalassemia (a form of anemia) caused by an imbalance in the coordinate production of α - and β -globins.

Listed below are problems commonly associated with gene therapy:

- I Localization of the introduced DNA to the target tissue and target location in the genome.
- II Difficulty in controlling the output of the introduced DNA.
- III Gene expression could be transient.
- IV Post-transcriptional modification could be missing.

Which of the above are illustrated by the findings on mice transfected with rabbit β -globin gene?

- A I, II and III only
 - B II and IV only
 - C III and IV only
 - D All of the above
39. In terms of containment (prevention of genetic pollution of closely-related wild plant or non-GM crop populations), which of the following is an advantage of chloroplast transformation (i.e. the introduction of foreign genes into chloroplasts of host plants) over nuclear transformation?
- A Chloroplasts are surrounded by a double membrane.
 - B There are no chloroplasts in pollen of most plant species.
 - C Chloroplasts are smaller than the nucleus.
 - D There is no DNA in chloroplasts.

40. Normal tomato plants have a gene that codes for an enzyme that softens tomatoes as they ripen.

The DNA sequence for this gene is as follows:

3' ... T T A G C C T T A ... 5',

Genetically engineered tomatoes ripen and soften more slowly because of the insertion of a transgene which reduces the amount of the softening enzyme produced.

Which of the following would be the mRNA sequence encoded by this transgene?

- A 5' – A U U C C G A U U – 3'
- B 5' – A A T C G G A A T – 3'
- C 5' – A A U C G G A A U – 3'
- D 5' – U U A G C C U U A – 3'

END OF PAPER